BMRC WEEK 8:
Environmental Emergencies

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Scenario #1

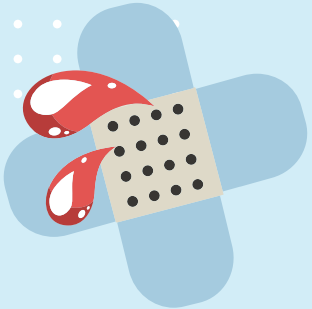
Dispatch to... “A roadwork site for a 24-year-old male with weakness after taking a fall”

What is your field impression?



Correct Answer:

Partial-thickness burns
and
mild heat exhaustion



MORE ABOUT..

- Definition: Heat exhaustion results from excessive heat and fluid loss, burns occur when skin contacts a hot surface.
- Signs & symptoms: Sweating, thirst, weakness, dizziness, rapid pulse, and painful red skin with blisters (partial-thickness burn)
- Potential progression: Heat exhaustion can progress to heat stroke, a life-threatening emergency involving central nervous system dysfunction, such as confusion, seizures, or loss of consciousness
- Why this patient is lower priority initially: BP 110/72, HR 104, RR 20, SpO_2 98%, and normal mental status indicate stable breathing and circulation, with mild heat-exhaustion symptoms
- Cooling: Move to shade, loosen excess clothing, and use a fan to reduce body heat. Offer small sips of water if fully alert and able to swallow safely.
- Burn treatment: Cool the burn with cool running water for up to 20 minutes, avoid ice and leave blisters intact. Cover with a clean, dry, sterile dressing.
- Note: Confusion or altered mental status suggests heat stroke and requires immediate emergency escalation. The burn is partial-thickness and approximately 1% TBSA.

Alameda County (ALCO) Protocols:

BURN PATIENT CRITERIA

1. **INTRODUCTION** -The intent of this policy is to transport patients with critical burns, who have a manageable airway, directly to a facility that is staffed and equipped to care for the medical needs of the patient, bypassing other receiving facilities. Minor to moderate burn patients will be transported to the closest, most appropriate receiving hospital.
2. **BURN PATIENT CRITERIA** (from the American Burn Association – Burn Unit Referral Criteria)
   - 2.1 Partial thickness burns greater than 10% total body surface area
   - 2.2 Moderate to severe burns that involve the face, hands, feet, genitalia, perineum, or major joints
   - 2.3 Full thickness burns in any age group
   - 2.4 Electrical burns, including lightning injury
   - 2.5 Chemical burns
   - 2.6 Burn injury in patients with preexisting medical disorders that could complicate management, prolong recovery, or affect mortality
   - 2.7 Inhalation injury
3. **DESTINATION**
   - 3.1 **Adult and Pediatric patients** who meet burn patient criteria 2.1-2.6 may be transported directly to an out-of-county burn center (see #5 below).
   - 3.2 Exceptions:
     - 3.2.1 **Potentially unmanageable airway** - (e.g. - soot in the mouth and/or nose, inhalation injury, etc.) transport to the closest trauma center.
     - 3.2.2 **Unmanageable Airway** - The patient requires intubation, and the paramedic is unable to intubate, and an adequate airway cannot be maintained with B.V.M. device, transport to closest basic E.D.
     - 3.2.3 **Patient meets Critical Trauma Patient Criteria** – "Mental Status & Vital Signs" - transport to the closest most appropriate designated trauma center
4. **OUT-OF-COUNTY TRANSPORT**
   - 4.1 Transporting medic ***must*** first contact out-of-county hospital to confirm bed availability. This can be done through the appropriate dispatch center or via phone or radio from the field
   - 4.2 Contact the Base Physician if medical consultation is needed
   - 4.3 Consider EMS Aircraft transport for land transport times greater than 45 minutes
   - 4.4 Give a brief report to the receiving facility including ETA

Out-of-County Burn Centers:

FACILITY	ADULT	PEDIATRIC	TRAUMA	HELIPAD	LOCATION	PHONE #
Santa Clara Valley Medical Center	YES	YES	YES	YES	751 S. Bascom Ave., San Jose	(408) 885-6666
UC Davis Medical Center	YES	YES	YES	YES	2315 Stockton Blvd., Sacramento	(916) 734-3636
St. Francis Memorial Hospital	YES	NO	NO	NO	900 Hyde Street, San Francisco	(415) 353-6255

BURN PATIENT CARE

Routine Medical Care
- Rescuer safety
- Assume airway/respiratory involvement
- Stop the burning process - ***DO NOT USE COLD PACKS***
- Assess for associated trauma

A. BASIC MANAGEMENT

1. **Rule out airway damage**
   - 1.1 Assess for thermal airway injury and smoke inhalation
   - 1.2 High flow oxygen is critical
   - 1.3 Be prepared for intubation
2. **Assess and expose**
   - 2.1 Assess ABCs
   - 2.2 Perform a mini neurological exam - level of consciousness
   - 2.3 Expose and examine the patient for other areas of burn
   - 2.4 Remove jewelry, but do not remove stuck clothing
3. **Start IVs**
   - 3.1 Two large bore IVs (for major burns)
4. **Give IV fluids**
   - 4.1 Any patient regardless of age with suspected 2nd and 3rd degree burns over 20% TBSA, should be given IV with Normal Saline at the below rate. Boluses are discouraged except in cases of shock.
     - → **Adult:** 500 cc/hour (80 drops per minute) or (1-2 drops every second)
     - → **Pediatric:**
       - ≤ 5 years old: 125 cc/hour (20 drops per minute) or (1 drop every 3 seconds)
       - 6-12 years old: 250 cc/hour (40 drops per minute) or (1 drop every 1-2 seconds)
       - ≥ 13 years old: 500 cc/hour (80 drops per minute) or (1-2 drops every second)
5. **Document severity and treat the pain**
   - 5.1 Estimate the severity of the burns using the "Rule of 9s" and "Rule of 1s." The Rule of 1s uses the patient's palm, including fingers, representing approximately 1% of total body surface area, to estimate burn size
   - 5.2 **Treat pain.** Pain management should be considered mandatory for moderate to severe burns. See Pain Management Policies – Adult (page 47) and Pediatric (page 76)
6. **Protect against hypothermia and infection - dress burns**
   - 6.1 Keep patient warm to prevent hypothermia (use sheets or blankets)
   - 6.2 Burns involving less than10% total body surface area (TBSA):
     - → Pour cool running water on the affected area for 20 minutes, if possible
       - Use available tap water (e.g., garden hose) for cooling, sterile water is not required. Do not delay transport to complete the full 20 minutes; cooling does not have to continue in transit.
     - → Apply a dry sterile dressing
   - 6.3 Burns involving greater than or equal to 10% TBSA:
     - → Apply a dry sterile dressing

BURN PATIENT CARE

7. **Elevate burned body parts - 30°**
8. **Address psychological needs**
   - 8.1 Be honest and compassionate
   - 8.2 Consider anxiolytics – **Contact Base Physician for midazolam**
9. **Maintain body temperature and observe for hypothermia**

B. ELECTRICAL BURNS

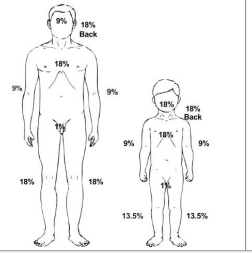
1. Turn off the power source if patient is still attached
2. See first responder defibrillation protocol if patient is unconscious and pulseless

C. TAR BURNS

1. Do not attempt to remove the tar
2. Cool with water
3. **Maintain body temperature and observe for hypothermia**

D. CHEMICAL BURNS

1. Remove clothing
2. **Liquid chemicals:**
   - →Flush **immediately** with copious amounts of tepid water for 15 - 20 minutes
3. **Dry chemicals:**
   - →Brush off as much as possible, then flush with copious amount of tepid water for 10 - 15 minutes
4. Identify chemical
5. Assess for associated respiratory burns

Rule of 9s	Rule of 1s
	Patient's palm and fingers are 1% TBSA 

BMRC First Aid Scope Protocol:

1. Ensure scene safety and move the patient away from the hot grill into shade
2. Call 911 yourself and request additional medical personnel as needed.
3. Assess the burn and identify the location, blisters, and approximate size
4. Cool the burn with cool running water, do not apply ice or pop blisters
5. Cover the burn with a clean, dry, sterile dressing
6. Cool the patient: rest in shade, loosen excess clothing, and use a fan
7. Offer small sips of water if fully alert and able to swallow safely
8. Check vital signs: BP,HR, and RR
9. Monitor for heat stroke: confusion, altered mental status, collapse, or seizures require immediate escalation
10. Reassess and hand off to EMS, reporting the burn and heat-related symptoms

Scenario #2

You're dispatched to a 20 YO female with difficulty breathing

What was your field impression?



Correct Answer:

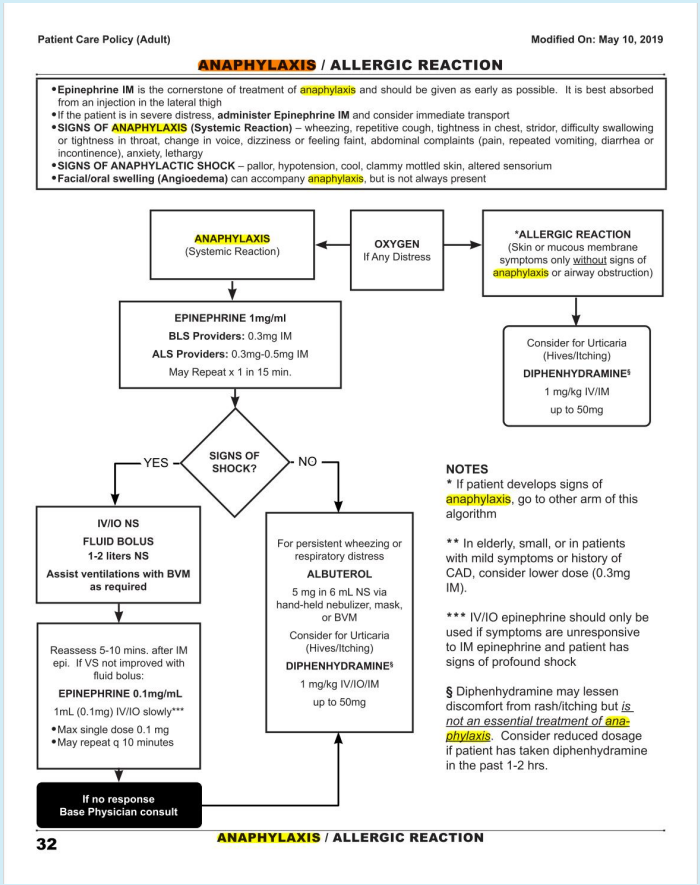
ANAPHYLAXIS



MORE ABOUT..

- Definition: Rapid, life-threatening allergic reaction, bee stings are a potential trigger
- S/S: Hives, itching, lip/throat swelling, hoarse voice, wheezing, difficulty breathing, dizziness, weak pulse, and hypotension
- Type of shock: Distributive (anaphylactic) shock – vasodilation and fluid leakage lower blood pressure and reduce overall organ perfusion
- Why this patient is high priority: BP 92/58, HR 120, RR 28, SpO_2 92%, throat tightness, and wheezing indicate respiratory compromise and possible shock
- Oxygen: An NRB mask provides high-concentration oxygen for a spontaneously breathing patient with hypoxemia & if breathing becomes inadequate, a BVM may be needed
- Key treatment: Epinephrine is the first-line treatment for anaphylaxis, oxygen is also supportive, not a substitute for epinephrine.
- Note: Symptoms can progress rapidly, even without hives so reassess ABCs 3-5mins

Alameda County (ALCO) Protocols:



Patient Care Policy (Adult) Modified On: May 10, 2019

ANAPHYLAXIS / ALLERGIC REACTION

- **Epinephrine IM** is the cornerstone of treatment of anaphylaxis and should be given as early as possible. It is best absorbed from an injection in the lateral thigh
- If the patient is in severe distress, **administer Epinephrine IM** and consider immediate transport
- **SIGNS OF ANAPHYLAXIS (Systemic Reaction)** – wheezing, repetitive cough, tightness in chest, stridor, difficulty swallowing or tightness in throat, change in voice, dizziness or feeling faint, abdominal complaints (pain, repeated vomiting, diarrhea or incontinence), anxiety, lethargy
- **SIGNS OF ANAPHYLACTIC SHOCK** – pallor, hypotension, cool, clammy mottled skin, altered sensorium
- **Facial/oral swelling (Angioedema)** can accompany anaphylaxis, but is not always present

ANAPHYLAXIS
(Systemic Reaction)

OXYGEN
If Any Distress

***ALLERGIC REACTION**
(Skin or mucous membrane symptoms only without signs of anaphylaxis or airway obstruction)

EPINEPHRINE 1mg/ml
BLS Providers: 0.3mg IM
ALS Providers: 0.3mg-0.5mg IM
May Repeat x 1 in 15 min.

Consider for Urticaria (Hives/Itching)
DIPHENHYDRAMINE§
1 mg/kg IV/IM
up to 50mg

SIGNS OF SHOCK?
YES
NO

IV/IO NS
FLUID BOLUS
1-2 liters NS
Assist ventilations with BVM as required

Reassess 5-10 mins. after IM epi. If VS not improved with fluid bolus:
EPINEPHRINE 0.1mg/mL
1mL (0.1mg) IV/IO slowly***
- Max single dose 0.1 mg
- May repeat q 10 minutes

If no response
Base Physician consult

For persistent wheezing or respiratory distress
ALBUTEROL
5 mg in 6 mL NS via hand-held nebulizer, mask, or BVM
Consider for Urticaria (Hives/Itching)
DIPHENHYDRAMINE§
1 mg/kg IV/IO/IM
up to 50mg

NOTES

* If patient develops signs of anaphylaxis, go to other arm of this algorithm

** In elderly, small, or in patients with mild symptoms or history of CAD, consider lower dose (0.3mg IM).

*** IV/IO epinephrine should only be used if symptoms are unresponsive to IM epinephrine and patient has signs of profound shock

§ Diphenhydramine may lessen discomfort from rash/itching but *is not an essential treatment of anaphylaxis*. Consider reduced dosage if patient has taken diphenhydramine in the past 1-2 hrs.

32 **ANAPHYLAXIS / ALLERGIC REACTION**

BMRC First Aid Scope Protocol:

1) Ensure scene safety and put on PPE
2) Call 911 and request ALS
3) Recognize anaphylaxis: hives, throat tightness, wheezing, and dizziness
4) Position for comfort and prevent unnecessary movement
5) Check vital signs: manual BP, pulse, and respiratory rate
6) Monitor continuously for worsening breathing or responsiveness
7) Prepare CPR/AED if the patient becomes unresponsive and is not breathing normally
8) Provide a handoff to EMS with symptoms, vital signs, and changes in condition

HYPOTHERMIA & HYPERTHERMIA

HYPOTHERMIA: Core body temperature drops below 95° F (35°C)

S/S: shivering, cold/pale skin, confusion, slurred speech, lethargy, bradypnea, bradycardia

Treatments:

- Move patient to a warm, dry environment
- Remove wet clothing and insulate with blankets
- Warm the core first (chest, neck, armpits)
- Monitor ABCs

HYPERTHERMIA: Core body temperature rises above 100.4°F (38°C)

S/S: heavy sweating, lethargy, headache, muscle cramps, confusion, seizures, tachypnea, tachycardia

Treatments:

- Move patient to a cool, shady environment
- Remove excess clothing
- Give small sips of cool water or an electrolyte drink only if **fully** alert and able to swallow
- Monitor ABCs

HEAT CRAMPS, HEAT EXHAUSTION, HEAT STROKE

CRAMPS

Painful muscle spasms caused by heat and fluid/electrolyte loss. Patient is usually alert. Treat with rest, cooling, and fluids/electrolytes if safe to drink.

EXHAUSTION

Excessive sweating and fluid loss causing weakness, dizziness, headache, nausea, and rapid pulse. Mental status is usually normal. Treat with cooling, rest, and fluids if alert and able to swallow.

STROKE

Life-threatening with signs of overheating, altered mental status, confusion, seizures, or unconsciousness. Skin is now be hot and dry. Activate EMS and cool rapidly immediately.

104F +

BURNS

SUPERFICIAL

Affect only the epidermis (outer skin layer). Skin is red, dry, painful, and has no blisters. Treat by cooling with cool running water and covering loosely.

PARTIAL

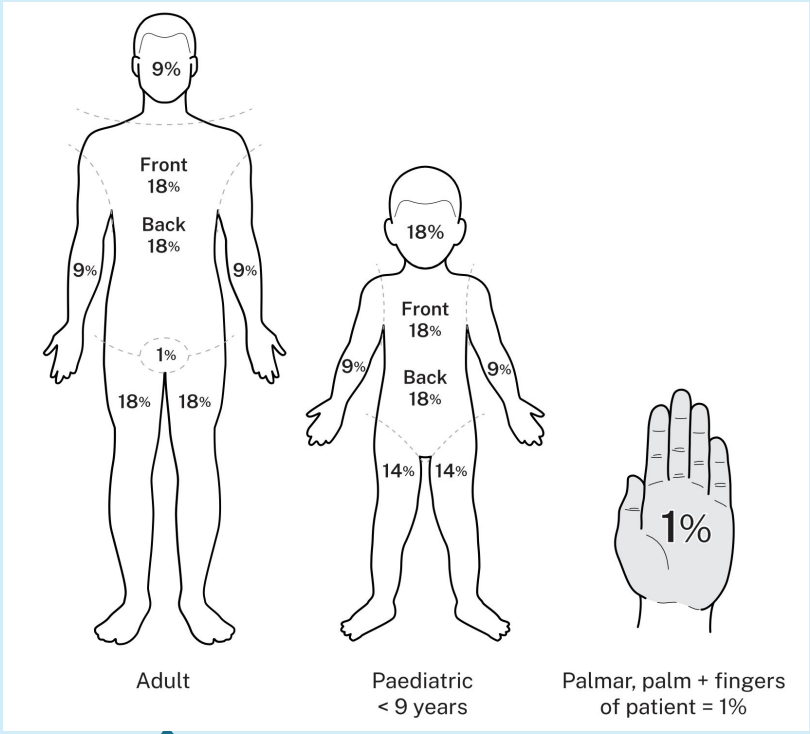
Affect the epidermis and part of the dermis. Skin is red, blistered, swollen, and very painful. Cool with clean, cool running water, cover with a sterile, non stick dressing, and do not pop blisters.

FULL

Destroy the epidermis and dermis and may extend into deeper tissues. Skin may look white, brown, black, leathery, or charred & the center may have little or no pain due to nerve damage. Activate EMS for serious burns, cover with a clean, dry dressing, prevent hypothermia, and monitor ABCs.

TIP: remove jewelry and loose clothing (not anything stuck to the skin), and assess for airway injury. Do not apply ice, butter, or ointments to serious burns.

RULE OF NINES & PALMAR METHOD



CROWD SYNDROME

This is a dangerous scenario that occurs when a large crowd becomes panicked or compressed, causing people to be crushed, trampled, or unable to breathe

- Ensure scene safety; watch for crowd movement and hazards
- Assess ABCs
- Treat life-threatening injuries, especially breathing problems and crush injuries, within EMT scope
- Move patients to a safe area when possible and request additional resources/MCI response if needed

THERMAL/TEMPERATURE INDUCED SHOCK

Shock caused by extreme heat or cold, which disrupts normal body function and circulation, reducing blood flow and oxygen delivery to vital organs.

- Remove the patient from the extreme temperature environment
- Assess ABCs and treat life threats
- Heat: Cool the patient and remove excess clothing
- Cold: Remove wet clothing and gently warm the patient
- Administer oxygen if indicated, prevent further heat loss, and transport promptly

LIGHTNING STRIKES & ELECTRIC SHOCK

Lightning: injury caused by a lightning strike, which can cause cardiac arrest, burns, and neurological damage

S/S: cardiac/ respiratory arrest, burns/ branching skin marks, confusion, seizures/ loss of consciousness, hearing/ vision problems, trauma

Treatments:

- Ensure scene safety, lightning can strike again!
- Assess ABCs
- Treat burns and other injuries
- Assess for spinal trauma and transport promptly

Electricity: injury caused when electrical current passes through the body, potentially damaging tissues and disrupting heart rhythm

S/S: entry/exit burns, cardiac arrest, muscle spasms, difficulty breathing, confusion/ loss of consciousness, pain, numbness/tingling of extremities

Treatments:

- Do not touch the patient until the electrical source is turned off
- Assess ABCs
- Assess for burns and associated trauma
- Monitor vital signs and transport promptly, even if external burns appear minor

DROWNING & SECONDARY DROWNING

Drowning: respiratory impairment caused by submersion or immersion in liquid

S/S: difficulty breathing/ respiratory arrest, coughing, choking/ gasping, low oxygen levels, AMS/ unconsciousness, vomiting, cardiac arrest

Treatments:

- Remove patient from water safely
- Assess airway and breathing
- Provide oxygen and ventilatory support as needed
- Prevent heat loss and transport

Secondary: delayed respiratory complications that can occur after a drowning or submersion incident

S/S: persistent/ worsening cough, difficulty breathing/ rapid breathing, chest tightness, fatigue/ unusual sleepiness, confusion/ AMS, low oxygen levels

Treatments:

- Assess ABCs, respiratory effort, and oxygen saturation
- Provide oxygen or ventilatory support as needed
- Transport for medical evaluation if symptoms are present or worsening

ALTITUDE-RELATED EMERGENCIES

HACE (high altitude cerebral edema): Swelling of the brain caused by exposure to high altitude and low oxygen levels

S/S: severe headache, confusion/ AMS, ataxia (loss of coordination), vomiting, decreased LOC

Treatments:

- Descend to a lower altitude as soon as safely possible
- Administer oxygen if available
- Keep the patient warm and minimize exertion
- Request evacuation and transport urgently

HAPE (high altitude pulmonary edema): Fluid buildup in the lungs caused by exposure to high altitude and low oxygen levels

S/S: SOB (especially at rest), rapid breathing & heart rate, persistent cough, possibly with pink/ frothy sputum, crackling lung sounds, cyanosis, severe fatigue/weakness

Treatments:

- Assess ABCs, respiratory effort, and oxygen saturation
- Provide oxygen or ventilatory support as needed
- Transport for medical evaluation if symptoms are present or worsening

DECOMPRESSION SICKNESS

THE BENDS

A condition in which dissolved nitrogen forms bubbles in the body when a diver ascends too quickly from deep water

S/S: Joint/ muscle pain, numbness/ tingling, dizziness/ weakness, difficulty breathing, confusion or paralysis, fatigue, skin rash

Treatments: administer 100% oxygen if possible, keep the patient at rest and monitor ABCs, transport for urgent medical evaluation, arrange hyperbaric oxygen treatment

BITES & STINGS



SNAKE BITES

Venomous: Inject venom that can damage tissue, nerves, or blood clotting.
Nonvenomous: Cause wounds and possible infection.
Rattlesnakes: May cause pain, spreading swelling, and bruising.
Care: Limit movement, remove tight jewelry, and transport promptly. No ice, tourniquets, cutting, or suction. Never handle the snake.



SPIDER BITES

Most spiders have venom, but most bites cause only mild pain, redness, and swelling.
Black widows: Venom can cause severe muscle cramps, abdominal pain, and sweating.
Care: Clean the bite, apply a wrapped cool compress, and seek care for severe symptoms.



TICK BITES

Often painless; some ticks spread illnesses such as Lyme disease.
Care: Grasp close to the skin with fine-tipped tweezers. Pull steadily without twisting, then clean the area.
Seek care if a rash or fever develops afterward.

Anaphylaxis

- A severe, life-threatening allergic reaction that can occur after a bite or sting.
- Signs: Throat swelling, wheezing, difficulty breathing, hives, dizziness, and low blood pressure.
- Care: Give IM epinephrine promptly per ALCO protocol, support breathing, request ALS, and transport rapidly.
- Prevention: Avoid known triggers and carry a prescribed EpiPen.

KAHOOT TIME!

THANK YOU!

